

Chapter 4

Limits at Infinity

§ Section: 2.6

4.1 Limits at Infinity

Definition:

Let f be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

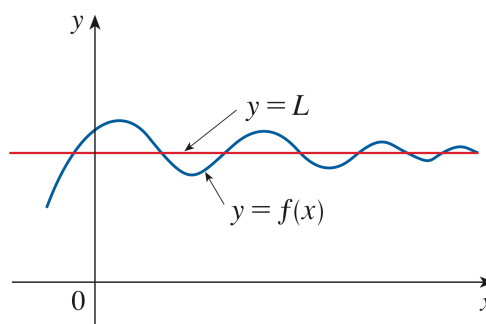
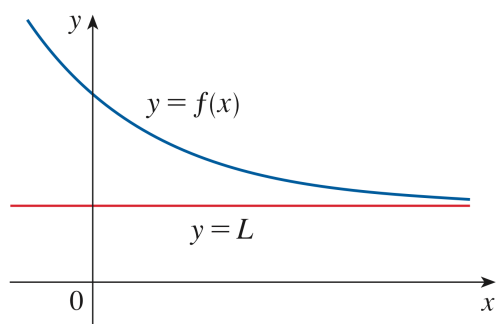
means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently large.

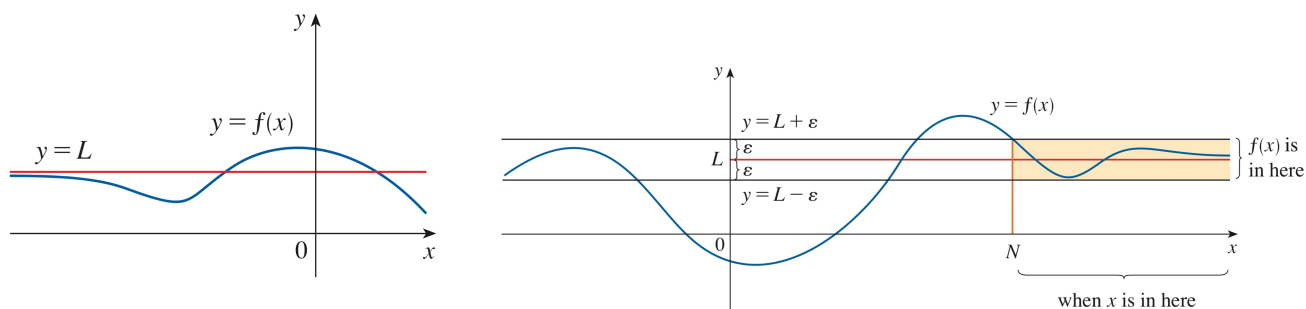
Definition:

Let f be a function defined on some interval $(-\infty, a)$. Then

$$\lim_{x \rightarrow -\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently large negative.



**Definition:**

The line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

Remark:

Limit Laws and the Squeeze Theorem apply to limits at infinity as well. Additionally, the limit at infinity of a composition involving a continuous function can be evaluated in the same way as with standard limits.

4.2 Compute Limits with Limit Laws

Theorem:

If $r > 0$ is a rational number, then

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$$

If $r > 0$ is a rational number such that x^r is defined for all x , then

$$\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0$$

Example 2.1

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0, \quad \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0, \quad \lim_{x \rightarrow -\infty} \frac{1}{x} = 0, \quad \lim_{x \rightarrow -\infty} \frac{1}{x^2} = 0,$$

$$\lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} = 0, \quad \lim_{x \rightarrow \infty} \frac{1}{\sqrt[3]{x}} = 0, \quad \lim_{x \rightarrow -\infty} \frac{1}{\sqrt[3]{x}} = 0, \text{ etc.}$$

Example 2.2

Find $\lim_{x \rightarrow \infty} \frac{2x^2 + 3x^{\frac{5}{3}}}{5x^2 + 4x - 2}$.

$$\lim_{x \rightarrow \infty} \frac{2 + 3 \frac{1}{x^{\frac{1}{3}}}}{5 + \frac{4}{x} - \frac{2}{x^2}} = \frac{2 + 0}{5 + 0 - 0} = \frac{2}{5}.$$

Example 2.3

Find $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 4x - 3}}{2x + 1}$.

$$\begin{aligned} &= \lim_{x \rightarrow -\infty} \frac{|x| \sqrt{1 + \frac{4}{x} - \frac{3}{x^2}}}{2x + 1} \\ &= \lim_{x \rightarrow -\infty} \frac{-x \sqrt{1 + \frac{4}{x} - \frac{3}{x^2}}}{2x + 1} \\ &= \lim_{x \rightarrow -\infty} \frac{-\sqrt{1 + \frac{4}{x} - \frac{3}{x^2}}}{2 + \frac{1}{x}} \\ &= -\frac{1}{2}. \end{aligned}$$

Example 2.4

Find all horizontal asymptotes of $y = f(x) = \frac{1}{\sqrt{x^2 + 2x + 2} - x}$

sol:

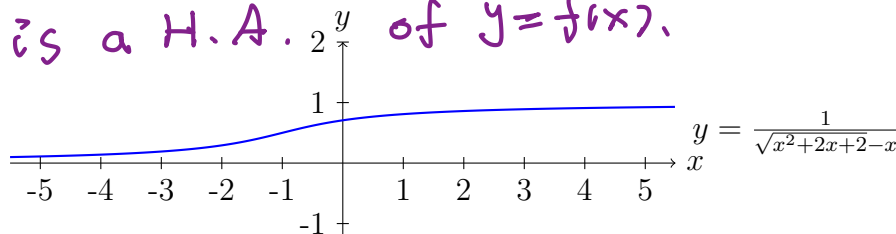
$$\begin{aligned} \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x^2 + 2x + 2} - x} = \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 2x + 2} + x}{(\sqrt{x^2 + 2x + 2} - x)(\sqrt{x^2 + 2x + 2} + x)} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 2x + 2} + x}{2x + 2} = \lim_{x \rightarrow \infty} \frac{|x| \sqrt{1 + \frac{2}{x} + \frac{2}{x^2}} + x}{2x + 2} \\ &= \lim_{x \rightarrow \infty} \frac{x \sqrt{1 + \frac{2}{x} + \frac{2}{x^2}} + x}{2x + 2} = \lim_{x \rightarrow \infty} \frac{\sqrt{1 + \frac{2}{x} + \frac{2}{x^2}} + 1}{2 + \frac{2}{x}} = 1. \end{aligned}$$

$\Rightarrow y=1$ is a horizontal asymptote of $y=f(x)$.

Sol 1:

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} \frac{-x \sqrt{1 + \frac{2}{x} + \frac{2}{x^2}} + x}{2x + 2} = \lim_{x \rightarrow -\infty} \frac{-\sqrt{1 + \frac{2}{x} + \frac{2}{x^2}} + 1}{2 + \frac{2}{x}} = 0$$

$\Rightarrow y=0$ is a H.A. of $y=f(x)$.



Sol 2: $\because \lim_{x \rightarrow -\infty} \sqrt{x^2 + 2x + 2} = \infty$ and $\lim_{x \rightarrow -\infty} -x = \infty \therefore \lim_{x \rightarrow -\infty} \sqrt{x^2 + 2x + 2} - x = \infty$

4.3 Compute Limits with the Squeeze Theorem

and $\lim_{x \rightarrow -\infty} \frac{1}{\sqrt{x^2 + 2x + 2} - x} = 0$.

Example 3.1

Find $\lim_{x \rightarrow -\infty} \frac{\sin x}{x}$

$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

Sol: $\because -1 \leq \sin x \leq 1 \therefore \frac{-1}{x} \geq \frac{\sin x}{x} \geq \frac{1}{x}$ for $x < 0$.

Moreover, $\lim_{x \rightarrow -\infty} \frac{-1}{x} = \lim_{x \rightarrow -\infty} \frac{1}{x} = 0$. Thus, $\lim_{x \rightarrow -\infty} \frac{\sin x}{x} = 0$ by the Squeeze Thm.

Example 3.2

Find $\lim_{x \rightarrow -\infty} \frac{[x^2]}{3x^2 - 2x}$

Sol: $\because x^2 - 1 < [x^2] \leq x^2$ for all x

$\therefore \frac{x^2 - 1}{3x^2 - 2x} < \frac{[x^2]}{3x^2 - 2x} \leq \frac{x^2}{3x^2 - 2x}$ for $x < 0$.

$\lim_{x \rightarrow -\infty} \frac{x^2 - 1}{3x^2 - 2x} = \frac{1}{3} = \lim_{x \rightarrow -\infty} \frac{x^2}{3x^2 - 2x}$

4.4 Compute Limits of Composition Functions

Example 4.1

Find $\lim_{x \rightarrow \infty} (\ln(1 + ax^2) - 2 \ln(2 + bx))$, where $a, b > 0$ are constants.

$$\lim_{x \rightarrow \infty} \ln \frac{(1 + ax^2)}{(2 + bx)^2} = \ln \left[\lim_{x \rightarrow \infty} \frac{1 + ax^2}{(2 + bx)^2} \right]$$

$$= \ln \left(\frac{a}{b^2} \right).$$

Example 4.2

Find $\lim_{x \rightarrow \infty} \cos \left(\frac{1}{x} \right) = \cos \left(\lim_{x \rightarrow \infty} \frac{1}{x} \right) = \cos 0 = 1$.

Example 4.3

Find $\lim_{x \rightarrow \infty} x \cdot \sin\left(\frac{1}{x}\right)$ $= \lim_{x \rightarrow \infty} \frac{\sin(\frac{1}{x})}{\frac{1}{x}} = \lim_{\substack{\text{Let } y = \frac{1}{x} \\ y \rightarrow 0^+}} \frac{\sin(y)}{y} = 1.$

4.5 Limits of Exponential Functions

Example 5.1

For $a > 1$, $\lim_{x \rightarrow -\infty} a^x = 0$. For $0 < a < 1$, $\lim_{x \rightarrow \infty} a^x = 0$.

Example 5.2

(a) Find $\lim_{x \rightarrow -\infty} \frac{1 + 2^x - 3^x}{e^x + 2 \cdot 3^x + 5} = \frac{1+0-0}{0+0+5} = \frac{1}{5}$

(b) Find $\lim_{x \rightarrow \infty} \frac{1 + 2^x - 3^x}{e^x + 2 \cdot 3^x + 5} = \lim_{x \rightarrow \infty} \frac{(\frac{1}{3})^x + (\frac{2}{3})^x - 1}{(\frac{e}{3})^x + 2 + 5 \cdot (\frac{1}{3})^x} = \frac{0+0-1}{0+2+0} = -\frac{1}{2}.$

4.6 Infinite Limits at Infinity

The notation $\lim_{x \rightarrow \infty} f(x) = \infty$ is used to indicate that the values of $f(x)$ become arbitrarily large as x becomes large. Similar meanings are attached to the following symbols:

$$\lim_{x \rightarrow \infty} f(x) = -\infty \quad \lim_{x \rightarrow -\infty} f(x) = \infty \quad \lim_{x \rightarrow -\infty} f(x) = -\infty$$

Definition:

Let f be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

means that for every positive number M there is a corresponding positive number N such that

$$\text{if } x > N \text{ then } f(x) > M$$

Example 6.1

$$\lim_{x \rightarrow \infty} x^n = \infty \text{ for all } n \in \mathbb{N}. \quad \lim_{x \rightarrow -\infty} x^n = \begin{cases} \infty, & \text{if } n: \text{ even} \\ -\infty, & \text{if } n: \text{ odd} \end{cases}$$

$$\lim_{x \rightarrow \infty} x^r = \infty \text{ for all } r \in \mathbb{Q} \text{ and } r > 0.$$

Example 6.2

$$\text{Find } \lim_{x \rightarrow -\infty} (x^4 - 3x) = \infty$$

$$\lim_{x \rightarrow \infty} x^4 - 3x = \lim_{x \rightarrow \infty} x^4 \left(1 - \frac{3}{x^3}\right) = \infty.$$

$\downarrow \quad \quad \downarrow$
 $\infty \quad \quad 1$

Example 6.3

$$\text{Find } \lim_{x \rightarrow \infty} \frac{-2x^{\frac{7}{3}} + 4x + 1}{x^2 - 3x^{\frac{1}{3}} + 2} = \lim_{x \rightarrow \infty} \frac{-2x^{\frac{7}{3}} + \frac{4}{x} + \frac{1}{x^2}}{1 - 3x^{-\frac{2}{3}} + \frac{2}{x^2}} = -\infty.$$

Example 6.4

For $a > 1$, $\lim_{x \rightarrow \infty} a^x = \infty$.

For $0 < a < 1$, $\lim_{x \rightarrow -\infty} a^x = \infty$.

Example 6.5

Find $\lim_{x \rightarrow 0^+} e^{\frac{x-1}{x}}$, $\lim_{x \rightarrow 0^-} e^{\frac{x-1}{x}}$, $\lim_{x \rightarrow \pm\infty} e^{\frac{x-1}{x}}$.

$$\text{Sol: } \because \lim_{x \rightarrow 0^+} \frac{x-1}{x} = -\infty \quad \therefore \lim_{x \rightarrow 0^+} e^{\frac{x-1}{x}} = 0$$

$$\because \lim_{x \rightarrow 0^-} \frac{x-1}{x} = \infty \quad \therefore \lim_{x \rightarrow 0^-} e^{\frac{x-1}{x}} = \infty$$

$$\because \lim_{x \rightarrow \pm\infty} \frac{x-1}{x} = 1 \quad \therefore \lim_{x \rightarrow \pm\infty} e^{\frac{x-1}{x}} = e^1 = e.$$

Exercise 6.1

Discuss $\lim_{x \rightarrow \infty} \frac{P(x)}{Q(x)}$, where $P(x)$ and $Q(x)$ are polynomials such that

(a) $\deg P(x) < \deg Q(x)$ 0

(b) $\deg P(x) > \deg Q(x)$ $\pm \infty$

(c) $\deg P(x) = \deg Q(x)$ $\frac{\alpha_n}{\beta_n}$
 $= n$